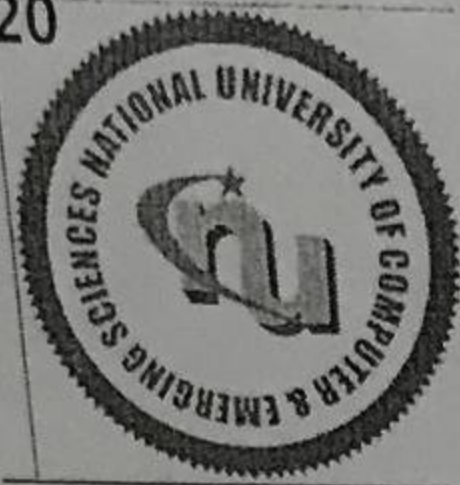


National University of Computer and Emerging Sciences, Lahore Campus

Course: Complex Variables and Transforms Course Code: MT-

220



Program:	BS (EE)	Semester:	Fall 2018
Duration:	1 hour	Total Marks:	25
Paper Date:	03-10-18	Weight	15%
Section:	All	Page(s):	1
Exam:	MID-I	Roll No:	

Instruction/Notes: Exchange of calculators or programmable calculators are not allowed at all.
Do all steps neat and clear.

Q #1 [6 marks, CLO-1]

Impedance measures the opposition of an electrical circuit to the flow of electricity. The total impedance in a parallel circuit is given by the formula.

$$Z_T = \frac{z_1 z_2}{z_1 + z_2}$$

- What is the total impedance of a circuit, Z_T , if $z_1 = -3 + 3i$ and $z_2 = -2 - 2i$.
- Represent the total impedance in polar form.
- Draw its graph in the complex plane.

Q#2 [3+4+1 mark, CLO-2]

Prove that the given function $u = \frac{1}{2} \log(x^2 + y^2)$ is harmonic. Also find the corresponding harmonic conjugate v and hence the original analytic function $f(z) = u + iv$.

Q#3 [4+4+3 marks, CLO-3]

- Find all complex values z satisfying the given equation.
 $\sin z = 2i$
- Prove the following identity using Euler's formula.
 $\sin x \cos^2 x = \frac{1}{4} (\sin 3x + \sin x).$
- Determine and sketch the set S of points in complex plane satisfying the given inequality.
 $|z - 1 - 3i| \leq 3$